

PROFILE

PhD in Natural Language Processing specialised in generating and evaluating **synthetic clinical data** under privacy constraints, now a Research Engineer at Arkhn building LLM-based clinical document generation with compact, self-hosted models. Work bridges large language models, the privacy–utility trade-off, and downstream medical tasks such as clinical entity extraction and ICD coding.

EXPERIENCE

- 2026 — **Research Engineer** Arkhn, Paris
LLM-based clinical document generation for Arkhn’s hospital assistant: agent architecture, structured patient-context with source traceability, human-in-the-loop workflows on compact, self-hosted models.
- 2022 — 26 **Doctoral Researcher (CIFRE)** Arkhn × Inria Almanach, Paris
Privacy-preserving synthetic clinical document generation with LLMs: facsimile documents, DP fine-tuning and iterative DPO, linkage/proximity privacy attacks, downstream evaluation on ICD-9 and semantic similarity.
- 2022 **NLP Engineer** Arkhn, Paris
NLP components for structuring clinical documents within Arkhn’s health-data platform, before transitioning to the industrial PhD.
- 2020 — 22 **NLP Data Scientist** 360 medics, Lyon
Low-resource clinical NER on French medical data: semi-automatic annotation, transformer deployment and latency optimisation, rule and classifier baselines.
- 2019 — 20 **Developer / Analyst** Orange, Nancy
Infrastructure test-automation: functional specifications, migration of the architecture to Kubernetes, continuous-integration setup.
- 2019 **Research Engineer** University of Glasgow, Computing Science
Natural language generation for social robotics — socially intelligent agents.
- 2015 — 19 **NLP Engineer** CNRS — ATILF, Nancy
Linguistic pipelines for ISTEEX / TermITH (terminology extraction, XML/TEI), technical lead on CoReA2D annotation tooling, Démonette web resource.

EDUCATION

- 2022 — 26 **PhD, Natural Language Processing** Sorbonne Université · defended 19 June 2026
- 2014 — 15 **M.Sc. Cognitive Science & NLP** Université de Lorraine
- 2009 — 12 **B.Sc. Cognitive & Computer Science** Université de Lorraine

SELECTED PUBLICATIONS

- 2025 **Synthetic Documents for Medical Tasks: Bridging Privacy with Knowledge Injection and Reward Mechanism** CL4Health
- 2024 **Generating Synthetic Documents with Clinical Keywords: A Privacy-Sensitive Methodology** CL4Health @ LREC-COLING ↗
- 2023 **Large Language Models as Instructors: A Study on Multilingual Clinical Entity Extraction** BioNLP @ ACL ↗
- 2023 **Annotate French Clinical Data Using Large Language Model Predictions** IEEE ICHI ↗
- 2023 **NL-Augmenter: A Framework for Task-Sensitive Natural Language Augmentation** NEJLT
- 2019 **Adapting NLG Methods to Social Robotics** NLP for Conversational AI @ ACL

SKILLS

- Full list — [Google Scholar](#) · [DBLP](#)
- ML / NLP LLM fine-tuning & alignment (Transformers, TRL, SFT / DPO), reward modeling, synthetic data generation, differential privacy, PyTorch, Lightning, spaCy
- LLM Agents LLM agents & orchestration (Claude SDK, LangGraph, Deep Agent), evaluation & observability (Weights & Biases, Langfuse)
- Engineering Python, Java, SQL, LaTeX / Typst, Docker, Kubernetes, vLLM, FastAPI, Git, CI/CD (GitHub Actions), GCP
- Spoken French (native), English (professional)